

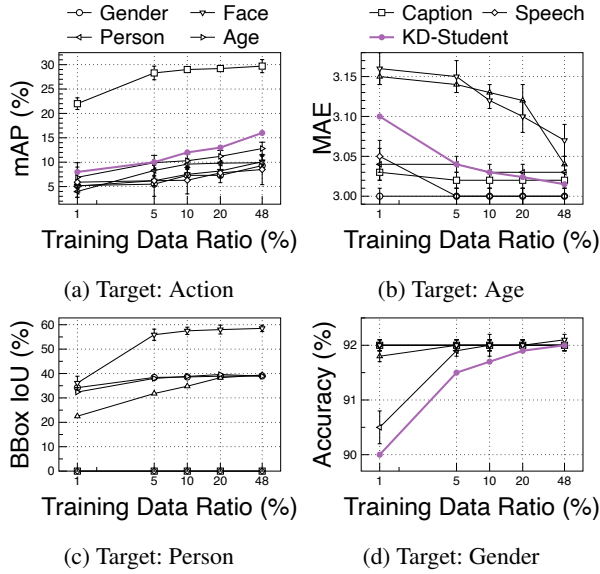


Figure 2: Performance of model links from different source models on four targets. KD-Student: student model trained via knowledge distillation on the target model.

Source	Action	Age	Face	Gender
IoU (%)	39.4 (± 0.1)	38.9 (± 0.1)	58.5 (± 1.3)	39.0 (± 0.1)
Corr.	0.123	0.042	0.244	-0.053

Table 2: IoU scores of model links targeted to the *Person* model and the Pearson correlations.

formance. The model link from *Face* to *Gender* achieves 92.1% accuracy. And for model links *Gender*-to-*Age* model achieves 3.0 MAE. Compared with student models trained via knowledge distillation on the target models, model links achieves higher prediction performance, especially when the amount of training data is small. But for speech recognition and video caption models, model links targeted to them cannot be effectively built and have around one WER score.

Discussions. We calculated the Pearson correlation coefficients between inference outputs of different models on the training split. For single-label and multi-label classification models, we used the index with the highest confidence as the label. For localization models, we checked whether the bounding box is empty, and assign 0 or 1 as the label. We used the regression scalar as the label and skipped the two sequence generation models. Tab. 2 shows the results of model links targeted to the *Person* model, and we can see a positive correlation between the model link performance and Pearson correlation coefficient.

Model link ensemble. For one target model, we have built multiple model links from different source models. Then we trained the ensemble models with all sources, using the same optimizer as model links and same hyperparameters. Tab. 3 shows the results on five target models, both model links and ensemble models were trained using

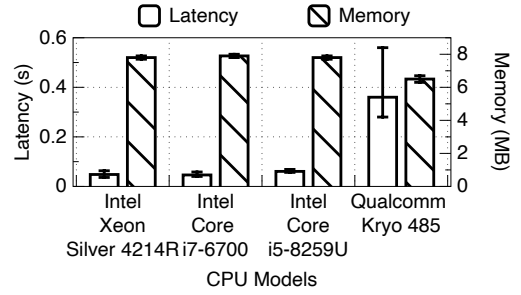


Figure 3: *MLink*'s overheads on servers and mobile phones.

all training samples (48% ratio). The model link ensemble outperforms every single source model. We can see there are two typical cases: dominance and mutual assistance. For *Action*, *Face*, *Person* targets, the *Caption*, *Person*, *Face* sources dominate the ensemble performance, respectively. But for *Age* and *Gender* targets, source models mutually assist and achieve performance improvement by ensemble.

Video Analytics with Model Links

We test *MLink* on 48-hour videos of 58 cameras in a smart building system and 48-hour videos of 10 cameras on a city traffic monitoring platform. We leveraged the first 10% in time of data for training model links and ensembles. We set the period length as one hour and use initial 1% data for profiling activation probabilities. For the counting models, the output accuracy is calculated by checking whether the absolute error of the predicted number is less than 0.5. The time costs of each ML model were the average inference time offline profiled by the training data. In the smart building system, the action/person/object models cost 30/44/60 ms per frame. In the traffic monitoring system, the traffic/person/vehicle models cost 55/66/70 ms per frame. The GPU memory costs of each ML model were the peak usage: 4.6 GB for person counting, 1.5 GB for action/traffic classification, and 3.7 GB for object/vehicle counting. We set the budget B as the maximal GPU memory allocated for ML models to evaluate how *MLink* improves the resource efficiency of multi-model inference. We treat every ML model's output accuracy equally and report their average output accuracy. Under GPU memory budget, the baseline "Standalone" simply selects the model with minimal average time cost. We repeated the scheduling experiments three times and reported the results in Tab. 4. Since the standard deviations are small (< 0.1), we did not present them for simplicity. In both systems, *MLink* outperforms alternatives in output accuracy. Compared with "Standalone", in the smart building system, *MLink* saves 66.7% inference executions, while preserving 94.1% output accuracy.

Overheads of *MLink*. We deployed *MLink* on four different devices (a cloud server, an edge server, a laptop, and a mobile phone) and tested its latency and memory footprint. As shown in Fig. 3, *MLink* only introduces negligible additional overheads.